

CaP-Agent0 Direct Synthesis

Perception



Handle alignment



Final state



```
grasps_green, scores_green = plan_grasp(...)
grasps_blue, scores_blue = plan_grasp(...)
```

```
target_vec = normalize_vector(pot_center_approx - handle_center)
for T_cam, score in zip(grasps, scores):
    T_world = pose_mat @ T_cam
    z_axis = T_world[:3, 2]
    y_axis = T_world[:3, 1]
    if abs(z_axis[2]) > 0.6: continue
    dot_dir = np.dot(z_axis, target_vec)
```

```
total_score = score + abs(y_axis[2]) * 2.5 + dot_dir * 1.5
```

```
move_to_joints_both(j0_lift, j1_lift)
```

❌ Failed

▲ Samples Contact-GraspNet candidates for each handle at runtime.

▲ Scores side-grasp candidates with axis and direction heuristics.

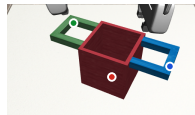
▲ Executes the planned lift, but the task condition remains false.

Sampled grasps and heuristic scoring do not form a reliable coordinated side grasp for the two handles.

Condensed policy code excerpts

RATS with learned-skills

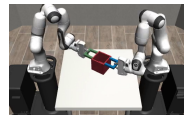
Perception



Handle alignment



Final state



```
pt0 = get_molmo_pt(rgb, "green handle..."); pt1 = get_molmo_pt(rgb, "blue handle...")
pot_pt = get_molmo_pt(rgb, "red main body...")
...
c0 = np.mean(pts0, axis=0); c1 = np.mean(pts1, axis=0)
c_pot = np.mean(pts_pot, axis=0)
```

```
z0 = c_pot[:2] - c0[:2]
z_axis0 = np.array([z0[0], z0[1], 0.0])
R0 = np.column_stack([x_axis0, y_axis0, z_axis0])
```

```
pre_grasp_pos0 = grasp_pos0 - z_axis0 * pre_dist
...
close_gripper_arm0(); close_gripper_arm1()
move_to_joints_both(lift_joints0, lift_joints1)
```

✅ Success

▲ Localizes handles and pot body, then computes 3D centers.

pick_moka_pot_by_handle

▲ Uses handle-to-body geometry to construct a side-pinch frame.

obb_aligned_side_pinch

▲ Executes side pre-approach, coordinated close, and lift.

horizontal_side_pinch

Selected skills guide policy writing for perception, side grasp, and lift.

Selected reusable skills (RATS)